PADHAKU
ICT – Class 6 · Class 6 ICT



CHAPTER 6

Being Future Ready - 01

NCERT Exercise Solutions

Introduction to Augmented Reality (AR)	4 Q
Introduction to Virtual Reality (VR)	1 Q
Self Assessment	1 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

⭐ Introduction to Augmented Reality (AR)

Activity 1 **Identify examples of Augmented Reality (AR) around you and write about five such examples.**

💡 **SAMPLE ANSWER**

AR means mixing virtual (computer-made) objects with the real world in real time. Five examples:

S.No.	Example
1	Live cricket score shown on the pitch on TV while real players play
2	Trying glasses frames on your own face in an online shopping app
3	Placing a sofa or table in your room with a furniture app before buying it
4	Google Maps Live View showing arrows on the real road while walking
5	Camera filters (stickers, masks) on a selfie, or seeing a 3D lion in your room with Google search

Activity 2 **Use Augmented Reality through Google Chrome (on a smartphone).**

💡 **SAMPLE ANSWER**

Steps:

- 1. On your Android phone, go to google.com or open the Google app.
- 2. Search for an animal, object or place (for example, “lion”).
- 3. If a 3D result is available, tap **‘View in 3D’**.

- 4. To see the 3D result in AR, tap **'View in your space'**.
- 5. Follow the on-screen instructions (move the phone slowly to scan the floor).

Result: The 3D lion appears inside your room through the phone camera. You can walk around it and see it from all sides.

Merge cube **Let us learn to make a paper Merge Cube ourselves.**

💡 SAMPLE ANSWER

- 1. Download the paper Merge Cube PDF from https://mergecube.com/paper-pdf
- 2. Print it on an 8.5″ × 11″ paper sheet.
- 3. Cut it along the **dashed black outline**.
- 4. Fold it along the **solid black lines** to give it a cube shape.
- 5. Put glue on all the flaps and stick them so that it becomes a cube.

Now scan the cube with a compatible app (for example, Merge Explorer) on an AR-compatible phone to hold virtual 3D objects in your hand.

Try **Now try these steps (Merge Explorer: install → Mr Body → Activity 1 → cube mode → hold the cube behind the phone) on other free applications.**

💡 SAMPLE ANSWER

Open another free Merge EDU app or activity, choose cube mode and hold the Merge Cube behind the phone.

Turn the cube to see the 3D model from all angles, and tap a part to see its name and details.

You can also try free AR apps such as ARLOOPA, 3DBear or Devar.

⭐ Introduction to Virtual Reality (VR)

Activity 3 **Make a Google Cardboard (VR headset). Link: https://www.designnews.com/makergadget-freak/make-your-own-cardboard-vr-goggles**

💡 SAMPLE ANSWER

Things needed: a cardboard sheet (e.g. a pizza box), two biconvex lenses (focal length about 45 mm), scissors/cutter, glue or tape, Velcro or a rubber band.

- 1. Print or draw the Google Cardboard template and stick it on the cardboard.
- 2. Cut out all the pieces, including the eye holes and the nose cut.
- 3. Fix the two lenses in the eye holes.
- 4. Fold the pieces into a box and glue/tape them.
- 5. Add Velcro or a rubber band to close the front flap that holds the phone.
- 6. Open a VR app or a 360° video on the phone, put the phone in the front and look through the lenses.

Result: You feel you are inside a 360° digital world – this is Virtual Reality.

(Take an adult's help when using the cutter.)

⭐ Self Assessment

1 **Fill the assessment sheet for these activities: grade by yourself (Student Assessment) and by one peer from your group (Peer Assessment).**

- **1. Identify AR examples around you and write about five.**
- **2. Use AR through Google Chrome.**
- **3. Make a Google Cardboard.**

Grades: D – Need teacher's help; C – Requires occasional peers' help; B – Can work independently; A – Is able to help others.

💡 **SAMPLE ANSWER**

For each activity, write your own grade and ask one friend from your group to write their name and grade. Be honest. Example:

S.No.	Activity	Student Assessment	Name of the Peer	Grade
1	Five AR examples	A	Riya	A
2	AR through Google Chrome	B	Riya	B
3	Google Cardboard	C	Riya	B

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**